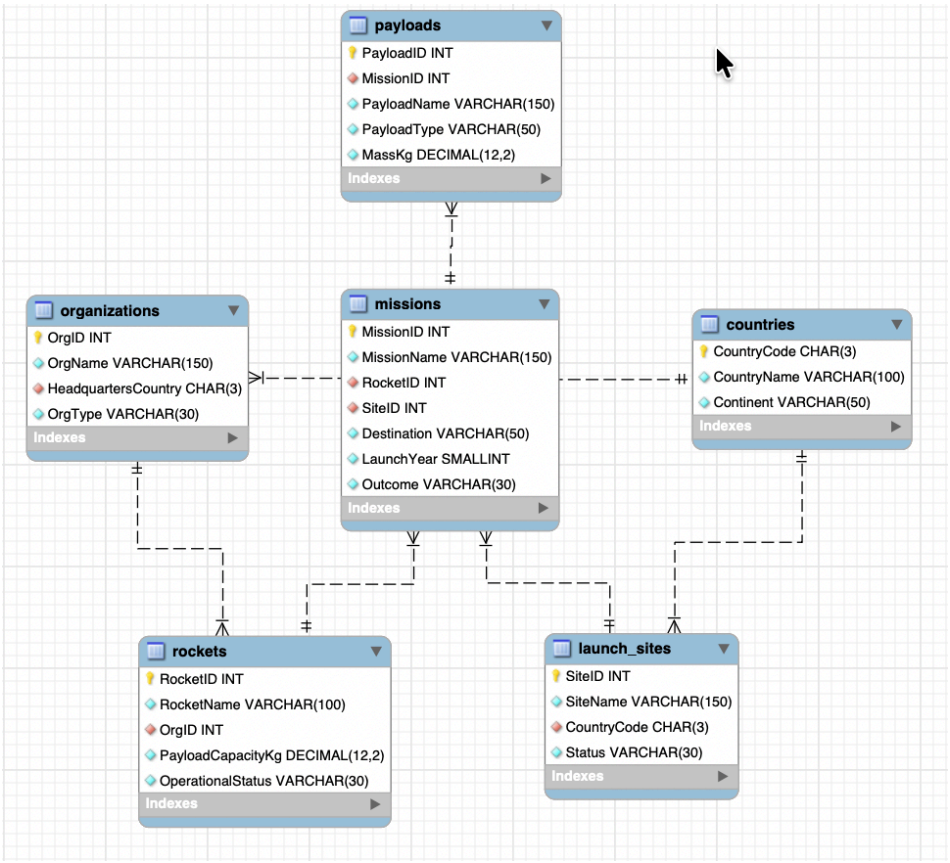
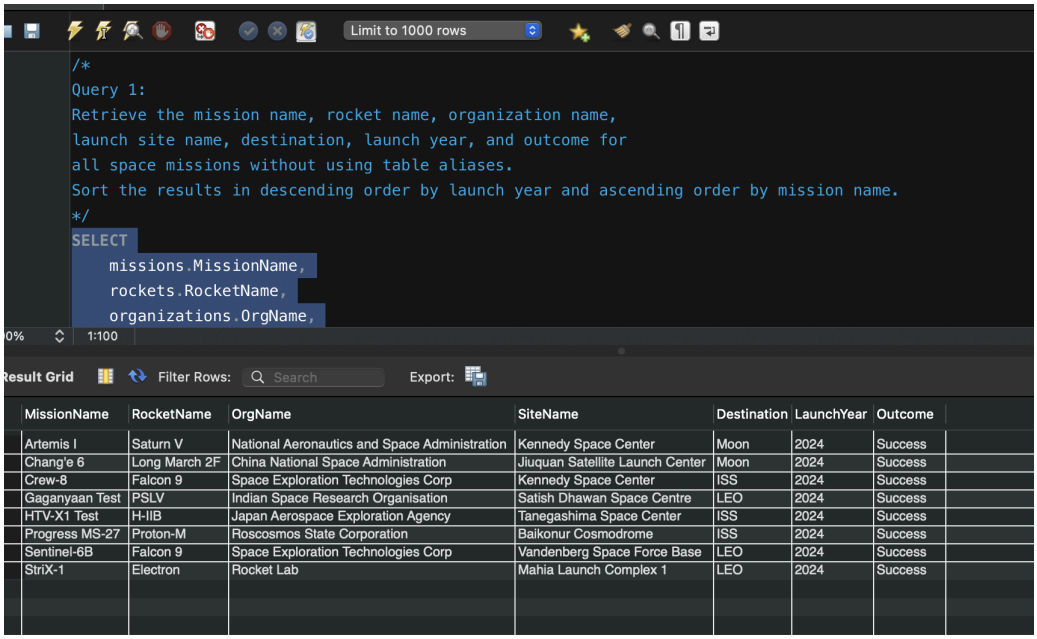
Luis Calvillo — Assignment 4

ER Diagram:

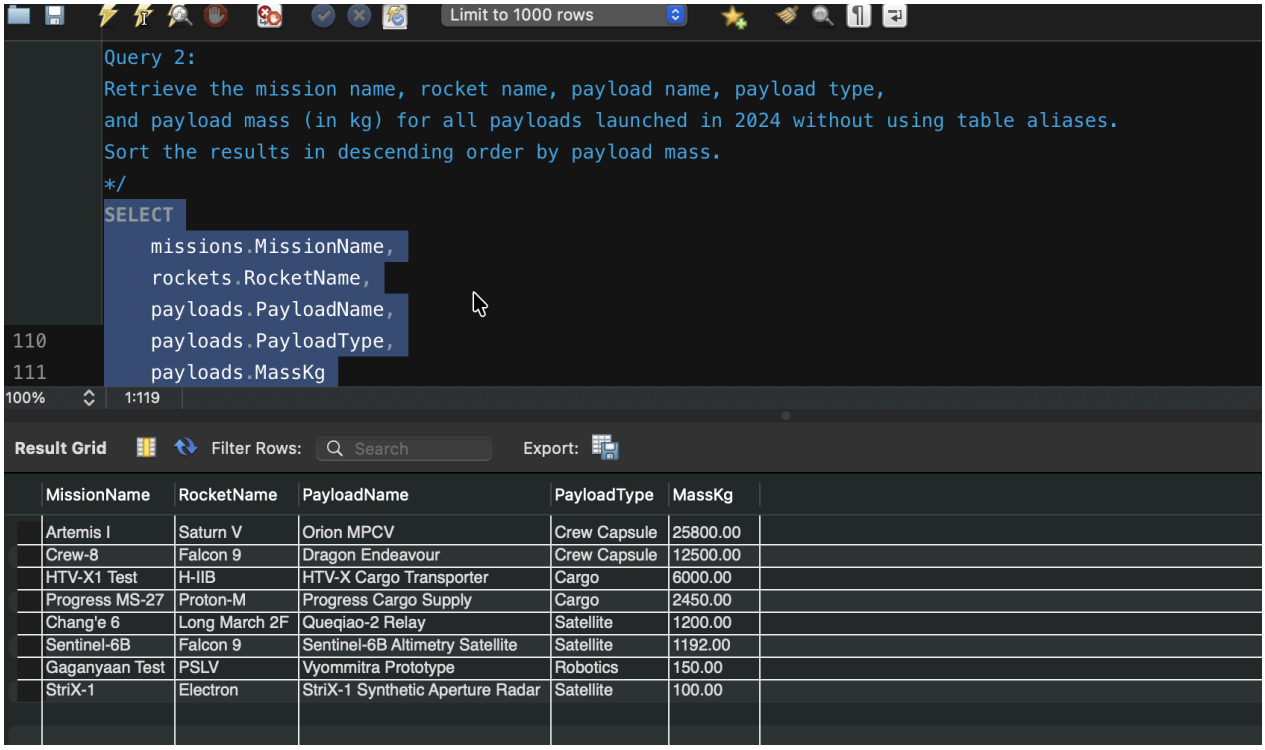


Query 1:



MissionName	RocketName	OrgName	SiteName	Destination	LaunchYear	Outcome
Artemis I	Saturn V	National Aeronautics and Space Administration	Kennedy Space Center	Moon	2024	Success
Chang'e 6	Long March 2F	China National Space Administration	Jiuquan Satellite Launch Center	Moon	2024	Success
Crew-8	Falcon 9	Space Exploration Technologies Corp	Kennedy Space Center	ISS	2024	Success
Gaganyaan Test	PSLV	Indian Space Research Organisation	Satish Dhawan Space Centre	LEO	2024	Success
HTV-X1 Test	H-IIB	Japan Aerospace Exploration Agency	Tanegashima Space Center	ISS	2024	Success
Progress MS-27	Proton-M	Roscosmos State Corporation	Baikonur Cosmodrome	ISS	2024	Success
Sentinel-6B	Falcon 9	Space Exploration Technologies Corp	Vandenberg Space Force Base	LEO	2024	Success
StriX-1	Electron	Rocket Lab	Mahia Launch Complex 1	LEO	2024	Success

Query 2:

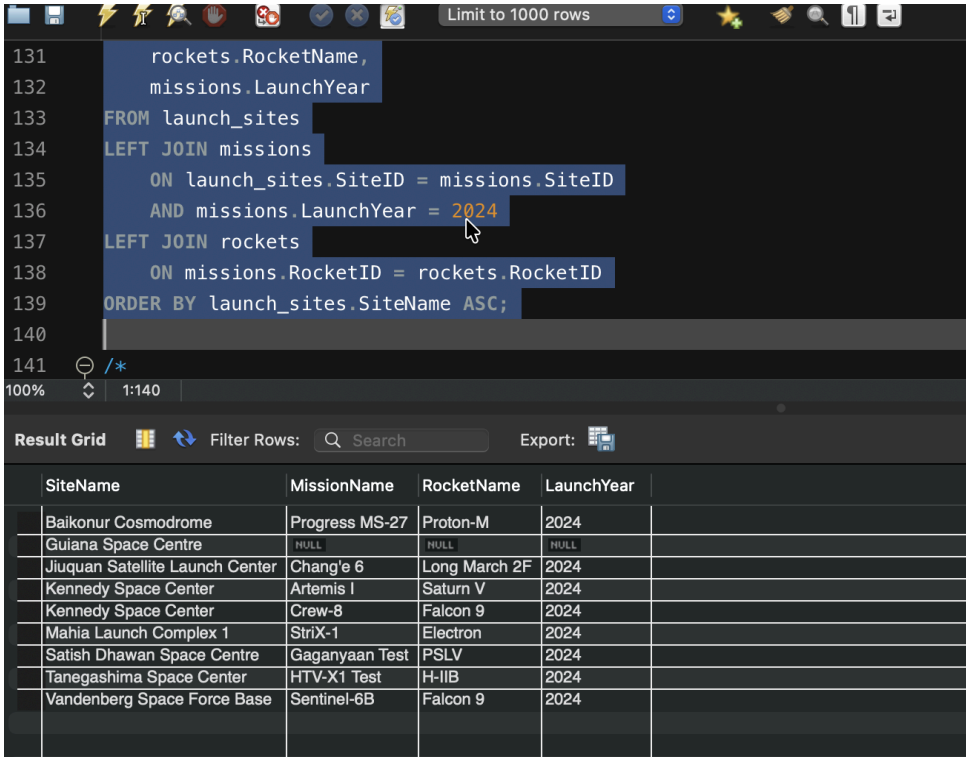


```
Query 2:
Retrieve the mission name, rocket name, payload name, payload type,
and payload mass (in kg) for all payloads launched in 2024 without using table aliases.
Sort the results in descending order by payload mass.
*/
SELECT
    missions.MissionName,
    rockets.RocketName,
    payloads.PayloadName,
    payloads.PayloadType,
    payloads.MassKg
```

Result Grid

MissionName	RocketName	PayloadName	PayloadType	MassKg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800.00
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500.00
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000.00
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450.00
Chang'e 6	Long March 2F	Queqiao-2 Relay	Satellite	1200.00
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Satellite	1192.00
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150.00
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Satellite	100.00

Query 3:



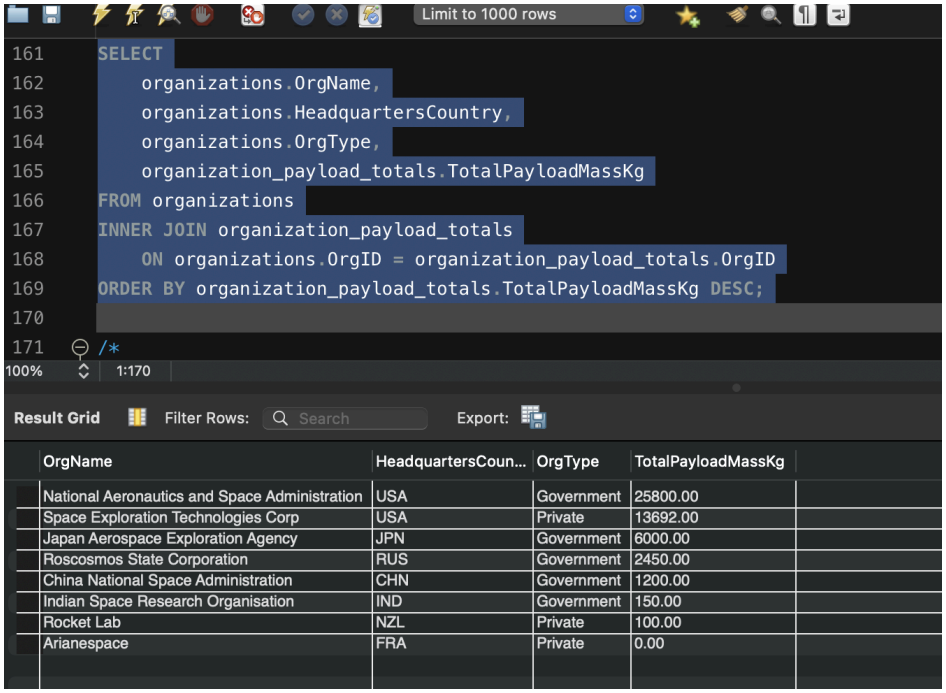
```
    rockets.RocketName,
    missions.LaunchYear
FROM launch_sites
LEFT JOIN missions
    ON launch_sites.SiteID = missions.SiteID
    AND missions.LaunchYear = 2024
LEFT JOIN rockets
    ON missions.RocketID = rockets.RocketID
ORDER BY launch_sites.SiteName ASC;

/*
```

Result Grid

SiteName	MissionName	RocketName	LaunchYear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March 2F	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

Query 4:



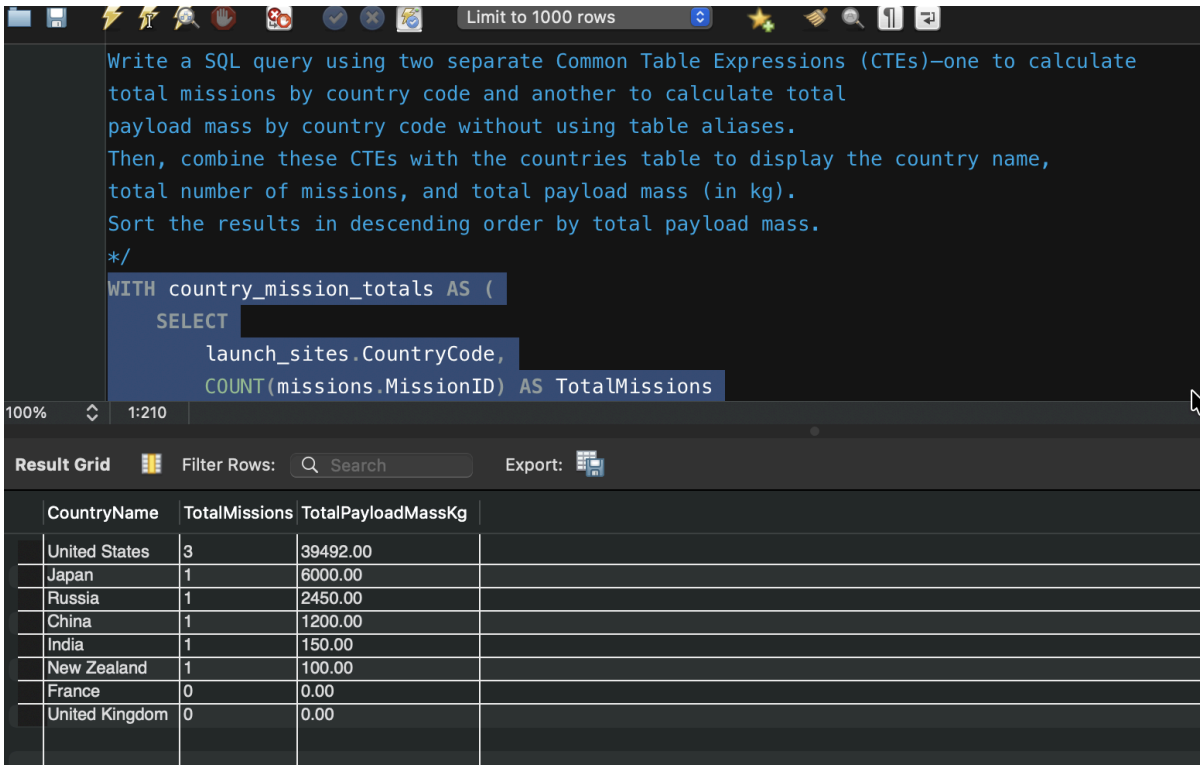
```
SELECT
    organizations.OrgName,
    organizations.HeadquartersCountry,
    organizations.OrgType,
    organization_payload_totals.TotalPayloadMassKg
FROM organizations
INNER JOIN organization_payload_totals
    ON organizations.OrgID = organization_payload_totals.OrgID
ORDER BY organization_payload_totals.TotalPayloadMassKg DESC;

/*
```

Result Grid

OrgName	HeadquartersCoun...	OrgType	TotalPayloadMassKg
National Aeronautics and Space Administration	USA	Government	25800.00
Space Exploration Technologies Corp	USA	Private	13692.00
Japan Aerospace Exploration Agency	JPN	Government	6000.00
Roscosmos State Corporation	RUS	Government	2450.00
China National Space Administration	CHN	Government	1200.00
Indian Space Research Organisation	IND	Government	150.00
Rocket Lab	NZL	Private	100.00
Arianespace	FRA	Private	0.00

Query 5



```
Write a SQL query using two separate Common Table Expressions (CTEs)—one to calculate
total missions by country code and another to calculate total
payload mass by country code without using table aliases.
Then, combine these CTEs with the countries table to display the country name,
total number of missions, and total payload mass (in kg).
Sort the results in descending order by total payload mass.
*/
WITH country_mission_totals AS (
    SELECT
        launch_sites.CountryCode,
        COUNT(missions.MissionID) AS TotalMissions
```

Result Grid

CountryName	TotalMissions	TotalPayloadMassKg
United States	3	39492.00
Japan	1	6000.00
Russia	1	2450.00
China	1	1200.00
India	1	150.00
New Zealand	1	100.00
France	0	0.00
United Kingdom	0	0.00